

Roman Urdu Medical Triage Classifier Comparison

Eight-page research summary | Supplied 10,000-record dataset

The study compares three classifiers for predicting triage levels from complaint representations and patient features. It uses the latest uploaded classifier settings as baselines, then evaluates additional settings under the same grouped test split. The language encoders remain frozen; training and tuning apply to the classifiers.

Records	Development	Final test	Target levels
10,000	8,001	1,999	1, 2 and 3

Main results

Model	Accuracy	Macro recall	Macro F1
Development-selected HGB	99.05%	0.9876	0.9888
Original fused HGB settings	99.15%	0.9878	0.9901
Random Forest comparison	99.35%	0.9950	0.9926

The selected system was chosen using development predictions, before the final test was evaluated. It did not exceed the strongest baseline on this test. Random Forest obtained the highest comparison score, but was not selected retrospectively from test results.

Data preparation and split

Twelve input fields and the target were retained. The constant Category field and five processing metadata columns were excluded. One invalid arrival entry was treated as missing; no records were removed in this run. Labels were not changed.

Records sharing the same normalized complaint or clinical concept were linked into 6,573 groups. Approximately 20% were reserved for testing. Three-fold screening and five-fold refinement used development groups only. Imputation, scaling, categorical encoding and PCA were fitted separately in each training fold.

Level	All records	Development	Test
1	3389	2712	677
2	4716	3773	943
3	1895	1516	379

Methods note: the supplied target labels were assigned by an AI model, as confirmed by the dataset provider. Scores measure agreement with those labels. The dataset has no level-4 examples; independent clinical validation was not part of this experiment.

Methods and final-model tables

SapBERT encodes clinical concepts with CLS pooling (768-D, 64-token limit). MPNet encodes concepts (768-D), and multilingual MiniLM encodes Roman Urdu complaints (384-D), both with a 128-token limit. Vectors are normalized. PCA 32, 64 and 128, and full dimensions, were tested. LR = Logistic Regression; HGB = Hist Gradient Boosting; RF = Random Forest. P/R/F1 are macro-averaged.

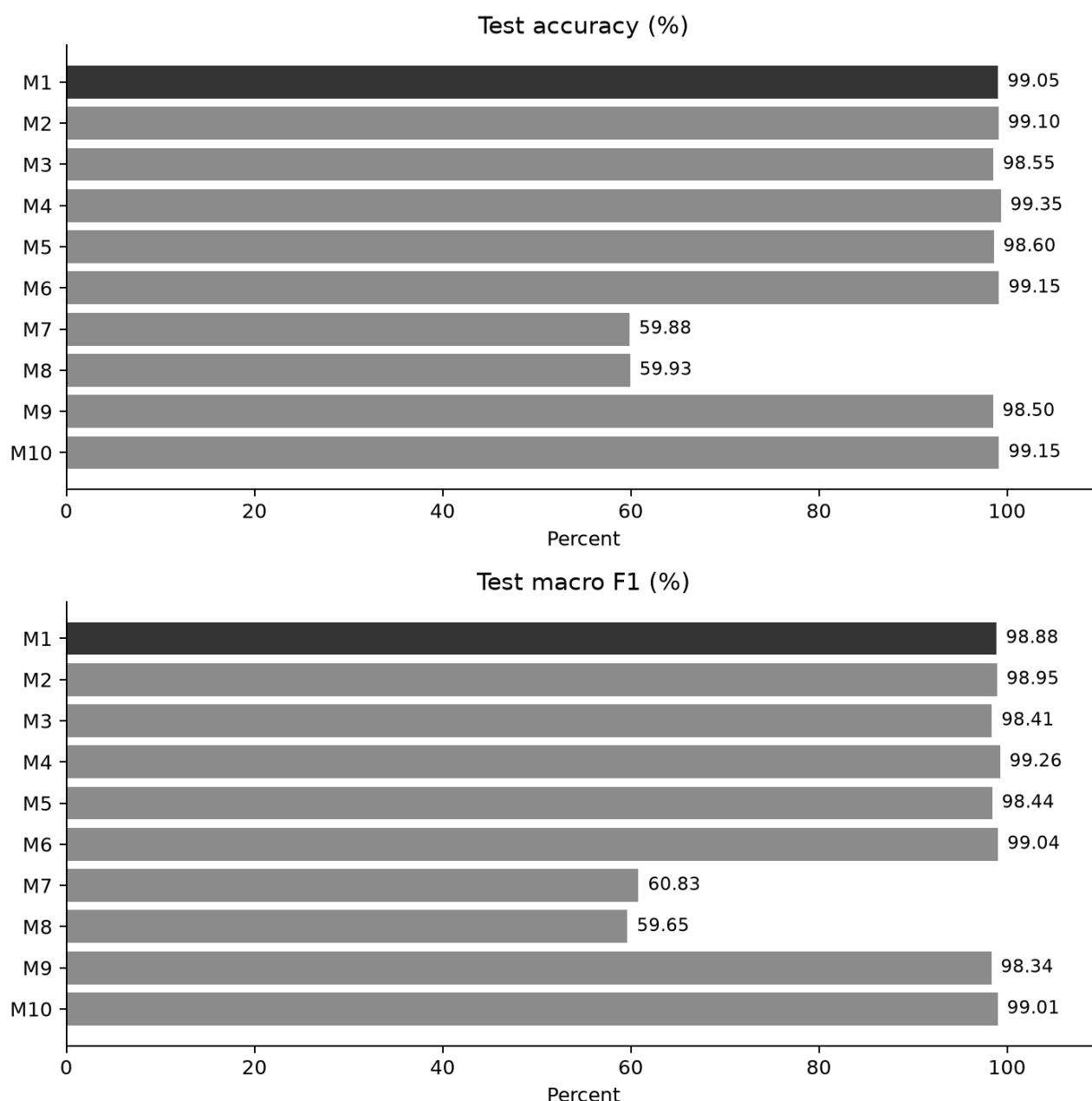
ID / configuration	Accuracy	Precision	Recall	F1
M1 Selected adjusted HGB	99.05%	0.9901	0.9876	0.9888
M2 HGB: Combined / SapBERT / 128D + derived	99.10%	0.9905	0.9885	0.9895
M3 LR: Combined / SapBERT / full	98.55%	0.9846	0.9835	0.9841
M4 RF: Combined / SapBERT / 64D	99.35%	0.9904	0.9950	0.9926
M5 LR: Patient features	98.60%	0.9807	0.9885	0.9844
M6 HGB: Patient features	99.15%	0.9932	0.9878	0.9904
M7 LR: Text only / SapBERT / 64D	59.88%	0.6410	0.5888	0.6083
M8 HGB: Text only / SapBERT / 64D	59.93%	0.6670	0.5705	0.5965
M9 LR: Combined / SapBERT / 64D	98.50%	0.9803	0.9866	0.9834
M10 HGB: Combined / SapBERT / 64D	99.15%	0.9924	0.9878	0.9901

Directional errors and agreement

ID	QWK	MAE	Under-triage	Over-triage
M1	0.9906	0.0095	9 (0.45%)	10 (0.50%)
M2	0.9911	0.0090	9 (0.45%)	9 (0.45%)
M3	0.9856	0.0145	16 (0.80%)	13 (0.65%)
M4	0.9936	0.0065	13 (0.65%)	0 (0.00%)
M5	0.9863	0.0140	22 (1.10%)	6 (0.30%)
M6	0.9915	0.0085	4 (0.20%)	13 (0.65%)
M7	0.4851	0.4302	344 (17.21%)	458 (22.91%)
M8	0.4671	0.4187	428 (21.41%)	373 (18.66%)
M9	0.9853	0.0150	22 (1.10%)	8 (0.40%)
M10	0.9915	0.0085	6 (0.30%)	11 (0.55%)

QWK = quadratic weighted kappa; MAE = mean absolute level error. Under-triage means predicting a higher numbered (less urgent) level; over-triage means predicting a lower numbered level. All ten rows use the same 1,999 test records.

Accuracy and F1 comparison



Read model IDs using the full configuration table on page 2. Text-only baselines are substantially lower than models that include patient features. Accuracy and macro F1 describe different aspects of performance, so both are retained.

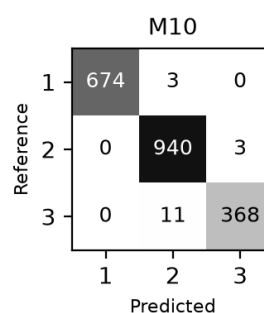
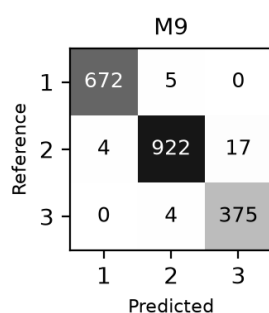
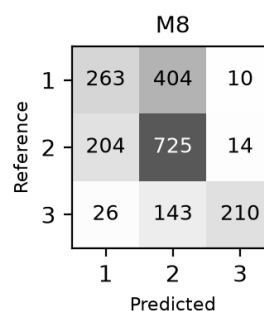
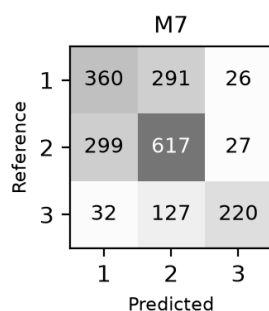
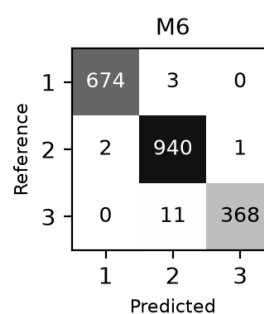
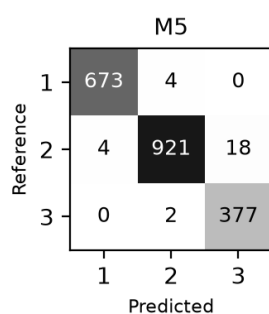
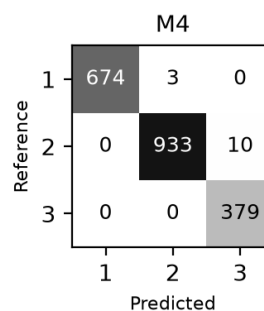
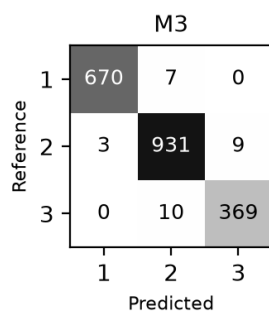
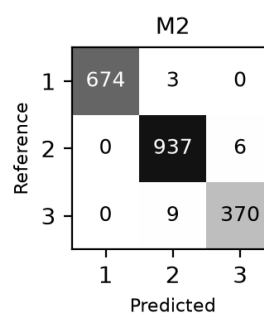
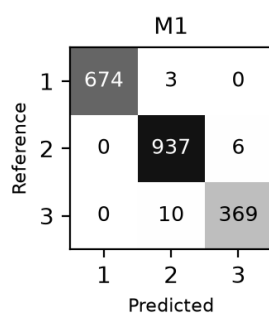
Selected-system uncertainty

Metric	Estimate	95% group-bootstrap interval
Accuracy	99.05%	98.62% to 99.42%
Macro F1	98.88%	98.38% to 99.35%

Intervals use 1,000 resamples of held-out groups. They describe sampling uncertainty within this dataset, rather than performance on a new hospital or independently labelled cohort.

Confusion matrices: all final models

Rows are reference levels; columns are predicted levels. Model IDs match page 2. Each matrix contains all 1,999 test records.



Per-class performance

Precision, recall and F1 are shown separately for every final model and triage level. Each model has support 677 / 943 / 379 for levels 1 / 2 / 3.

Model	Level	Precision	Recall	F1	Support
M1	1	1.0000	0.9956	0.9978	677
M1	2	0.9863	0.9936	0.9900	943
M1	3	0.9840	0.9736	0.9788	379
M2	1	1.0000	0.9956	0.9978	677
M2	2	0.9874	0.9936	0.9905	943
M2	3	0.9840	0.9763	0.9801	379
M3	1	0.9955	0.9897	0.9926	677
M3	2	0.9821	0.9873	0.9847	943
M3	3	0.9762	0.9736	0.9749	379
M4	1	1.0000	0.9956	0.9978	677
M4	2	0.9968	0.9894	0.9931	943
M4	3	0.9743	1.0000	0.9870	379
M5	1	0.9941	0.9941	0.9941	677
M5	2	0.9935	0.9767	0.9850	943
M5	3	0.9544	0.9947	0.9742	379
M6	1	0.9970	0.9956	0.9963	677
M6	2	0.9853	0.9968	0.9910	943
M6	3	0.9973	0.9710	0.9840	379
M7	1	0.5210	0.5318	0.5263	677
M7	2	0.5961	0.6543	0.6239	943
M7	3	0.8059	0.5805	0.6748	379
M8	1	0.5335	0.3885	0.4496	677
M8	2	0.5700	0.7688	0.6546	943
M8	3	0.8974	0.5541	0.6852	379
M9	1	0.9941	0.9926	0.9933	677
M9	2	0.9903	0.9777	0.9840	943
M9	3	0.9566	0.9894	0.9728	379
M10	1	1.0000	0.9956	0.9978	677
M10	2	0.9853	0.9968	0.9910	943
M10	3	0.9919	0.9710	0.9813	379

The selected system made 19 errors: nine under-triage and ten over-triage errors. Its class-specific results should be read alongside the aggregate scores, particularly for level 1.

Development search and model selection

The search completed 136 three-fold configurations and 22 five-fold refinements. It varied classifier parameters, class balancing, embedding representation, PCA size, derived features and text weight. Probability combinations and small class multipliers were selected using out-of-fold development predictions.

Five-fold configuration	Accuracy	Macro F1
HGB: Combined / SapBERT / 128D + derived [ebf1fb]	99.28%	0.9917
HGB: Combined / SapBERT / 64D + derived [0f3df5]	99.25%	0.9914
HGB: Combined / MPNet / 64D + derived [a55cdf]	99.24%	0.9912
HGB: Combined / MiniLM / 32D + derived [74be92]	99.24%	0.9912
HGB: Combined / MiniLM / 64D + derived [e905b9]	99.23%	0.9909
HGB: Combined / MiniLM / 128D + derived [8e1de6]	99.21%	0.9908
HGB: Combined / SapBERT / 32D + derived [f8d183]	99.19%	0.9907
HGB: Combined / MiniLM / 32D + derived [2a938b]	99.20%	0.9906
RF: Combined / SapBERT / 64D [37444b]	99.09%	0.9894
RF: Patient features [38acfc]	99.04%	0.9888
HGB: Patient features [96313a]	98.87%	0.9870
LR: Combined / SapBERT / full [bbd4c7]	98.46%	0.9837
LR: Combined / SapBERT / full [a3b46c]	98.38%	0.9829
LR: Combined / SapBERT / full [6b4606]	98.34%	0.9825
LR: Combined / SapBERT / full [254c1f]	98.31%	0.9822
LR: Combined / SapBERT / full [245e9d]	98.31%	0.9822
LR: Combined / SapBERT / full [92b5a9]	98.26%	0.9813
LR: Combined / SapBERT / full [99ada6]	98.16%	0.9803
LR: Patient features + derived [4cf94f]	98.14%	0.9800
HGB: Text only / MiniLM / 64D [a1ff41]	64.05%	0.6304
LR: Text only / MiniLM / full [e5e4ef]	61.48%	0.6148
RF: Text only / MiniLM / 64D [a7d542]	61.15%	0.6007

Selected settings

SapBERT concepts reduced to 128 PCA components, combined with patient features and three derived quantities: pulse pressure, heart-rate/systolic-pressure ratio, and a mean-pressure proxy. HGB uses 250 iterations, learning rate 0.1, 15 maximum leaves, L2 regularization 1 and balanced sample weights. Class probability multipliers are 0.8 / 1.0 / 0.8. This produces 153 input features.

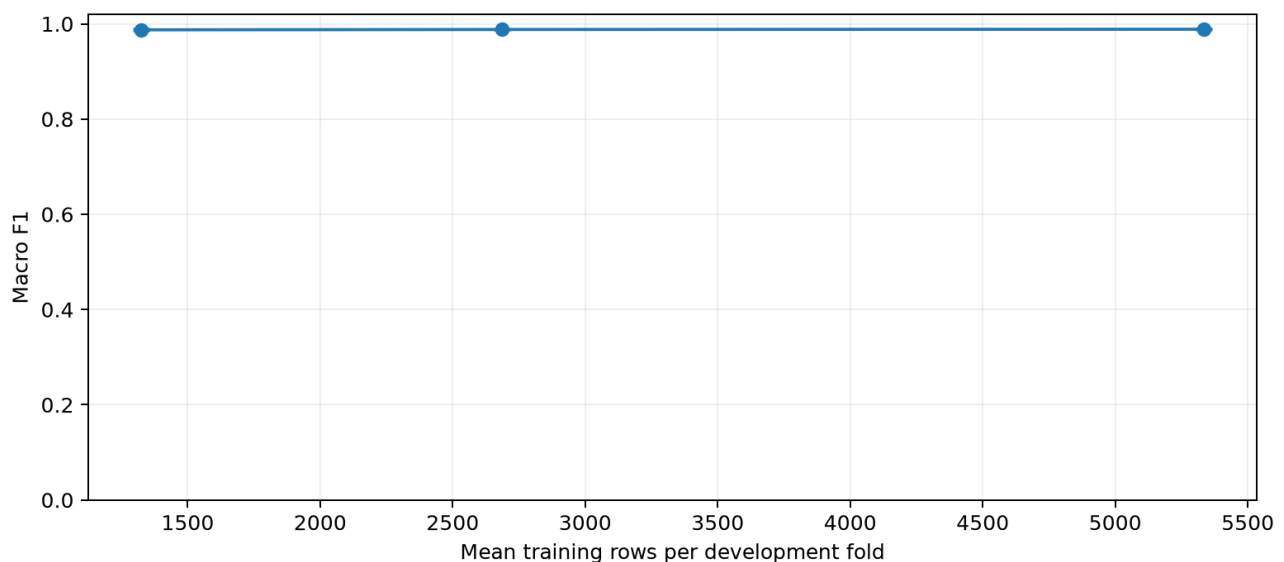
The six-character suffix identifies each configuration in the saved CV tables. The complete parameter records and all 136 screening results remain in the accompanying repository, avoiding dozens of near-repeated rows here. Development selection scores may be optimistic; final results use the separate test set.

Feature checks and learning behaviour

Development sensitivity test	Accuracy	Macro F1
Omit ECG_Status	97.48%	0.9738
Omit ECG_Status, Mode_of_Arrival	97.43%	0.9737
Omit Gender	98.95%	0.9878

These are development-only HGB feature-removal checks. The dataset contains strong relationships between patient features and labels; removing ECG, alone or together with arrival, reduces performance more than removing gender. These tests do not establish causal relationships.

Diagnostic	Training fraction	Accuracy	Macro F1
learning curve	0.25	98.90%	0.9874
learning curve	0.5	98.98%	0.9881
learning curve	1.0	99.03%	0.9887
shuffled training labels	1.0	42.42%	0.3241



Training-label shuffling is a negative control; learning curves compare group-based subsets of development training data. All diagnostic decisions remain separate from final-test selection.

Reproducibility

Split seeds: 42 for held-out groups and 123 for development folds. The original input checksum, encoder revisions, configuration JSON, full-precision metrics and verification record accompany the code. The saved experiment artifacts do not replace the deployed translation or inference pipeline.

Historical and literature comparisons

Historical experiments use different records, four target levels and different split protocols. Their scores provide context and are not controlled estimates of improvement on the new dataset. All historical final-model rows are retained below.

Earlier model / inputs	Accuracy	Recall	F1
SBERT LR / text only / full 768	58.09%	0.6058	0.5485
SBERT HGB / text only / full 768	57.43%	0.4720	0.4798
SBERT RF / text only / full 768	55.65%	0.5001	0.5108
SBERT LR / text only / pca 64	55.65%	0.5742	0.5178
SBERT HGB / text only / pca 64	56.98%	0.4658	0.4734
SBERT RF / text only / pca 64	54.32%	0.4649	0.4653
SBERT LR / fusion / full 768	82.48%	0.8176	0.7868
SBERT HGB / fusion / full 768	82.93%	0.7417	0.7697
SBERT RF / fusion / full 768	72.73%	0.6278	0.6437
SBERT LR / fusion / pca 64	82.26%	0.8162	0.7828
SBERT HGB / fusion / pca 64	80.71%	0.7464	0.7727
SBERT RF / fusion / pca 64	80.71%	0.7041	0.7306
SapBERT text 768+logreg	37.26%	0.4191	0.3576
SapBERT text 768+hgb	40.57%	0.3165	0.3130
SapBERT text 768+rf	42.92%	0.3338	0.3308
SapBERT fused 768+logreg	73.11%	0.7054	0.6837
SapBERT fused 768+hgb	71.70%	0.6101	0.6112
SapBERT fused 768+rf	65.09%	0.5188	0.5124
SapBERT text 64+logreg	34.43%	0.3982	0.3327
SapBERT text 64+hgb	39.15%	0.3016	0.2809
SapBERT text 64+rf	44.81%	0.3461	0.3374
SapBERT fused 64+logreg	72.64%	0.7019	0.6793
SapBERT fused 64+hgb	73.11%	0.6203	0.6329
SapBERT fused 64+rf	76.42%	0.6430	0.6602
SapBERT structured+logreg	73.11%	0.7044	0.6860
SapBERT structured+hgb	72.64%	0.5959	0.5971
SapBERT structured+rf	73.11%	0.6339	0.6554

Published comparison: Model / Accuracy / Recall / F1

Model [1]	Accuracy	Recall	F1
TF-IDF + Logistic Regression	0.750	0.988	0.544
TF-IDF + Random Forest	0.751	0.964	0.582
BiLSTM	0.746	0.846	0.670
CNN	0.723	0.787	0.667
Current selected HGB (this study)	0.9905	0.9876	0.9888

[1] Seo et al. (2025). Artificial intelligence for severity triage based on conversations in an emergency department in Korea. Scientific Reports 15, 16870. DOI: 10.1038/s41598-025-99874-0, Table 2. Published results use binary KTAS and ten-fold CV; averaging and task differ from the present three-class macro metrics. These are separate models, not a CNN-BiLSTM hybrid.